Optimisation landscape

Problem / Method	Solver / Algorithm	Software package[1] Open-source Commercial	Optimization reach Optimization-only Design-Decision-integrated	Characteristics
1. MILP - Single/multi objectives - **Linear objective** functions - (Conflicting) objectives - Mixed-integer variables - **Feasibility** constraints	**Linear programming (LP)** : Simplex; Branch-and-Bound/Cut (LP relaxation). MOO: scalarisation: weighted aggregation; ε-constraint; ideal-point/min-max; Pareto-front generation.	ORTools, HiGHS, SCIP, CBC, Choco-solver Gurobi, CPLEX, COPT, FICO Xpress	**MOO:** multi-objective optimisation for performance. Requires a-posteriori MCDA for evaluation; no direct best-fit	Highly scalable and fast. Strictly linear structure ⇒ global optimality guaranteed. Often not suitable for complex real-world engineering and scheduling problems. Mainly suitable for teaching and introductory use.
2. MINLP (MIP) - Single/multi objectives - **Non-linear objective** functions - (Conflicting) objectives - Mixed-integer variables - **Feasibility** constraints	**Constraint programming (CP):** constraint propagation. MOO: transformed CP or hybrid MILPs.	ORTools, SCIP BARON, AntiGone, Gurobi*, CPLEX*	**MOO:** multi-objective optimisation for performance. Requires a-posteriori MCDA for evaluation; no direct best-fit	Rigid; problem-class specific; requires exact math formulation. Guarantees objective optimality, but not decision-best-fit.
	Evolutionary heuristics: NSGA family, MOEA/D, SPEA, and BRKGA; population search and Pareto-front approximation.	Pymoo, DEAP	**MOO:** multi-objective Pareto optimality. Requires a-posteriori MCDA for evaluation; no direct best-fit	Black-box; Pareto methods cover only part of real-world problems. No guarantee of decision-best-fit (dominance in objective space only).
3. IMAP - Single/multi performance - **(Non)-linear** system **performance** functions - **(Non)-monotonic** multi-actor **preference** functions - (Conflicting) preferences - All type of variables - **Feasibility & acceptability** constraints	**Preference-Performance programming (PP):** Integrated Maximisation of Aggregated Preferences (IMAP). **Intergenerational heuristics** PFM-guided evolutionary search: IGS-GA family and IGS-SI. **Reinforcement Learning** PFM-guided sequential learning: IMAP-RL	PREFERENDUS	**MODO:** multi-objective integrated and aggregated preference optimisation, Directly solves for 'best-fit' design-decision point; pure support for multi-criteria and multi-actor design & decision-making (MCDM).	Design/decision-valid: integrates performance and preferences. Preferences explicitly modelled from performance and/or attributes. Yields a best-fit solution in aggregated weighted-preference design-decision space. Flexible for highly constrained, multi-actor real-world problems (incl. preference-based constraint handling). Also able to solve (non)linear SODOs.

[1] Among the most relevant
* Limited (mainly quadratic)